

# VG2392S240X0M2 Wireless Module Hardware Specifications

V1.0



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## 1. Overview

VG239 2 S240 X 0M 2 series wireless module is an external power amplifier ( PA ) and low noise amplifier ( LNA ) designed based on SEMTECH's SX1281 high-performance wireless transceiver chip , and a long-distance 2.4G LORA two-way wireless transceiver module. SX1281 is an ultra-long-distance communication wireless transceiver chip that supports LORA spread spectrum in the 2.4 GHz frequency band. It has the characteristics of high linearity and strong anti-interference. In order to further adapt to the complex application environment of the product and achieve a longer communication distance, the module integrates high-efficiency FEM devices. Compared with ordinary 2.4G wireless modules, its larger transmitting power and higher receiving sensitivity make it have better communication link budget capability.

The module integrates all RF-related functions and devices. Users can use this module to easily develop wireless solutions and wireless IoT devices with stable performance and high reliability without in-depth understanding of RF circuit design.

### Features:

- Long-distance 2.4G transceiver communication
- Module integrated PA, LNA amplifier
- High receiving sensitivity
- Programmable transmit power , the maximum transmit power can reach 27dBm
- At the same time support LoRa ®, FLRC, (G)FSK and other modulation methods
- Programmable data transfer rate

**Application:**

- Home Automation and Appliances
- IIoT asset management and security
- Logistics Tracking App
- Radio Controlled Toys and Drones
- Smart Agriculture
- medical insurance

## 2. Electrical Characteristics

| Parameter           | Description                                                           | Remark                                                       |
|---------------------|-----------------------------------------------------------------------|--------------------------------------------------------------|
| Power Supply        | 3.1 ~ 3.6V                                                            | Typically 3.3V                                               |
| Frequency Bands     | 2.4GHz                                                                | 2400MHz-2500MHz                                              |
| Output Power        | 27dBm                                                                 | The output power of the SX1281 chip is set to 0dBm           |
| Data Rate           | 125 ~ 2000Kbps@FSK<br>260 ~ 1300Kbps@FLRC<br>0.476 ~ 202Kbps@LoRa _ _ | Programmable configuration                                   |
| RF Modulation       | LoRa <sup>®</sup> , FLRC, (G)FSK                                      | Recommended LoRa , FLRC                                      |
| Receive sensitivity | -137dBm                                                               | LORA, SF12, BW=203kHz, CR=4/5                                |
| receive bandwidth   | 300 ~ 2400kHz @FSK<br>300 ~ 1200kHz @FLRC<br>203 ~ 1625kHz @LoRa      | Programmable configuration                                   |
| TX Current          | 370mA                                                                 | Transmit Power = 27dBm                                       |
| RX Current          | 17 mA                                                                 | LoRa BW=203KHz                                               |
| sleep Current       | < 18uA _                                                              | SX1281 is configured to enter sleep while TXEN=0, RXEN=0     |
| driver interface    | SPI                                                                   | Standard 4-wire SPI, SPI clock: <=10MHz, CPOL = 0 , CPHA = 0 |
| Antenna impedance   | 50 ohms                                                               |                                                              |

|                           |                                                      |                                                                                                                     |
|---------------------------|------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------|
| Antenna connection method | IPEX-1 seat, stamp half hole,<br>Onboard PCB Antenna | default onboard PCB antenna, if you need IPEX-1 socket or stamp half hole,<br>Need to modify the selection resistor |
| storage temperature       | -40 °C ~ +125 °C                                     |                                                                                                                     |
| Operating temperature     | -40°C ~ + 85°C                                       | Industrial grade                                                                                                    |
| Size                      | 16.0x24.0mm                                          |                                                                                                                     |

### 3. Pin Diagram

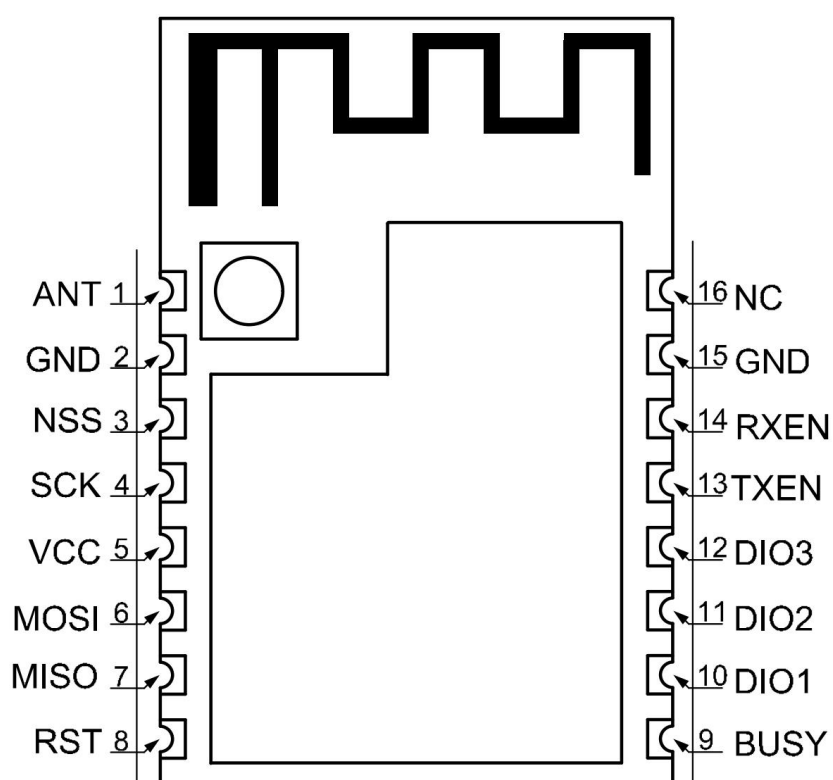


Figure 1-1 Top view

## 4.Pin Description

| Number | Name | Type         | Description                                                                                                   |
|--------|------|--------------|---------------------------------------------------------------------------------------------------------------|
| 1      | ANT  | I/O          | Antenna external interface, matching 50Ω, need to adjust the internal jump selection resistance of the module |
| 2      | GND  | power supply | land                                                                                                          |
| 3      | NSS  | I            | SPI interface SPI chip select                                                                                 |
| 4      | SCK  | I            | SPI interface SCLK clock input                                                                                |
| 5      | VCC  | power supply | Positive power supply                                                                                         |
| 6      | MOSI | I            | SPI interface MOSI data input                                                                                 |
| 7      | MISO | O            | SPI interface MISO data output                                                                                |
| 8      | RST  | I            | Reset signal, active low                                                                                      |
| 9      | BUSY | O            | Chip working status indication, busy status                                                                   |
| 10     | DIO1 | I/O          | Directly connected to the chip DIO1 digital I/O pin, software configurable function                           |
| 11     | DIO2 | I/O          | Directly connected to the chip DIO2 digital I/O pin, software configurable function                           |
| 12     | DIO3 | I/O          | Directly connected to the chip DIO3 digital I/O pin, software configurable function                           |
| 13     | TXEN | I            | Module PA control pin, TXEN=1 during transmission; RXEN=0, TXEN=0 during sleep; RXEN=0                        |
| 14     | RXEN | I            | Module LNA control pin, TXEN=0;RXEN=1 when receiving, TXEN=0;RXEN=0 when sleeping                             |
| 15     | GND  | power supply | land                                                                                                          |
| 16     | NC   | - -          | Internal suspension                                                                                           |

## 5. Hardware design guide

### 5.1. Application circuit

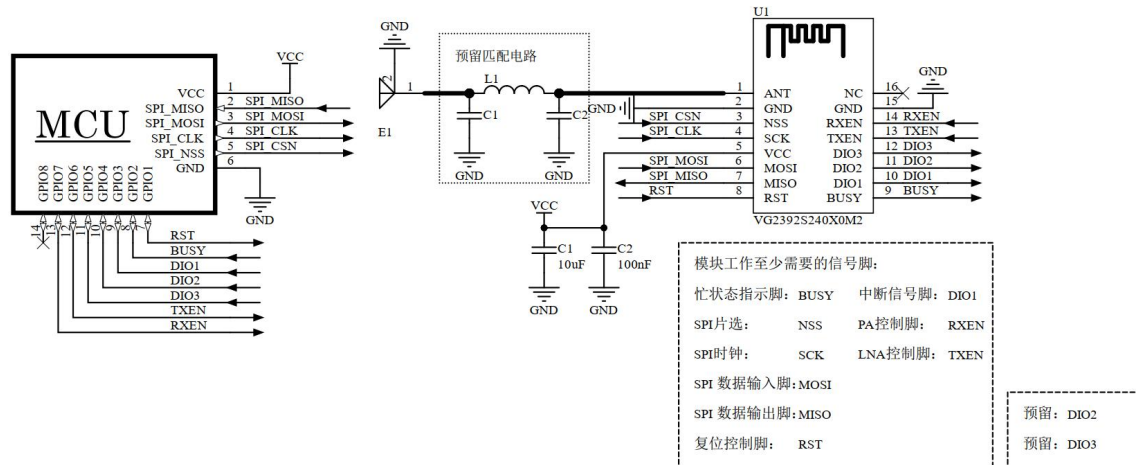


Figure 5-1 Programming development hardware connection ( stamp hole external antenna )

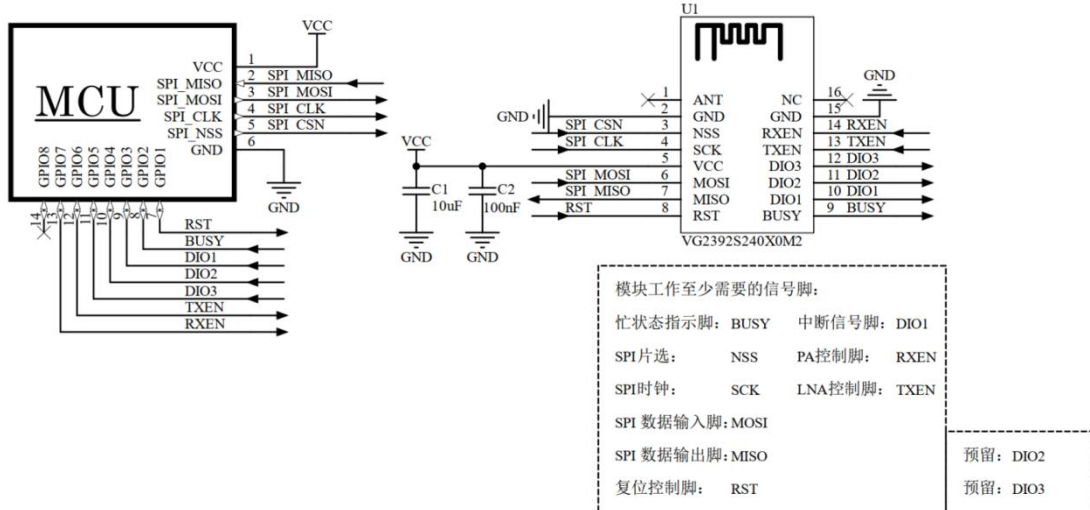


Figure 5- 2 Programming development hardware connection ( on-board PCB antenna )

### 5. 2. Power supply design

1. Please pay attention to the correct connection of the positive and negative poles of the power supply, and ensure that the power supply voltage is within the recommended power supply voltage range. If it exceeds the



maximum allowable power supply range of the module, the module will be permanently damaged; the filter capacitor of the module power supply pin should be as close as possible to the module power supply pin.

2. In the power supply system of the module, the excessive ripple may be coupled to the line that is easily interfered by the wire or the ground plane, such as the sensitive signal line such as the antenna, feeder, clock line, etc., which may easily cause the radio frequency performance of the module to deteriorate, so We recommend using LDO as the power supply for the wireless module.

3. When selecting the LDO voltage regulator chip, it is necessary to pay attention to the heat dissipation of the power supply and the driving capability of the LDO stable output current; considering the long-term stable operation of the whole machine, it is recommended to reserve more than 50% of the current output margin.

4. It is best to use a single LDO for the module to supply power; if a DC-DC power supply chip is used, an LDO must be added behind as the isolation of the module power supply to prevent the noise of the switching power supply chip from interfering with the working performance of the radio frequency.

5. If the communication line between the MCU and the module uses a 5V level, a 1K-5.1K resistor must be connected in series ( not recommended, there is still a risk of damage ) .

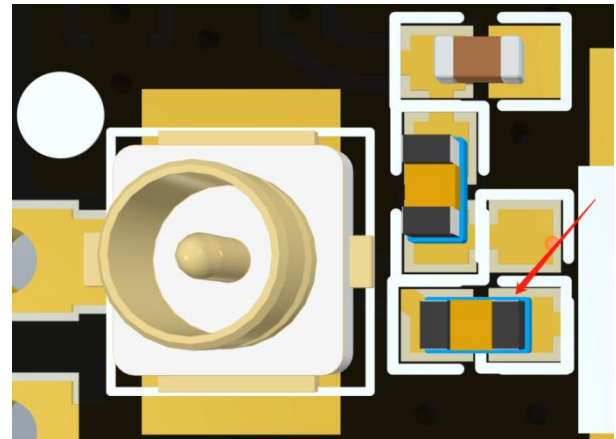
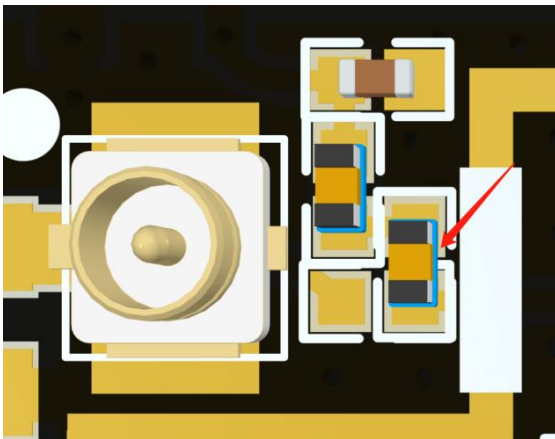
6. The RF module should be kept away from high-voltage devices as far as possible, because the electromagnetic waves of high-voltage devices will also have a certain impact on the RF signal.

7. High-frequency digital wiring, high-frequency analog wiring, and high-current power supply wiring should be avoided under the module as much as possible. If it is necessary to pass under the module, the wiring should be placed on another layer of the PCB bottom plate where the module is placed, and ensure that the module is under the module. The copper is well grounded.

## **5.3. Antenna Design and Guidance**

### **5.3.1. Selection of external antenna and PCB antenna**

module factory default is to select the on-board PCB antenna path. If you need to use an external antenna, you need to jump the transfer resistor to the external antenna path, as shown in the following figure:



5- 3 Onboard PCB Antenna Jump Resistor Connection

5- 4 IPEX-1 Seat or Stamp Half Hole Jumper Resistor Connection

### 5.3.2. Guidelines for bends in RF lines and RF trace

the RF output interface of the module is selected in the form of a stamp hole , a 50ohm characteristic impedance trace is used to connect the antenna on the backplane PCB during design. It should be noted that the RF traces on the backplane PCB should be as short as possible. The 2.4G signal is more sensitive to the trace length. It is recommended that the longest trace length should not exceed 10 mm, and the trace width should be kept continuous; Right angles, circular arcs are recommended.

|                                                      |  |
|------------------------------------------------------|--|
| The first recommended way of turning the RF traces   |  |
| Second, the recommended way of turning the RF traces |  |
| Bad way of turning RF traces , not recommended       |  |

In order to ensure that the RF trace on the backplane is 50 ohms, the following parameters can be adjusted according to different board thicknesses. The following simulation values are for reference only.

|                                |                                                                                     |
|--------------------------------|-------------------------------------------------------------------------------------|
| RF traces use 20mil line width | thickness is 1.0mm , the spacing between ground copper and traces is 5.3mil         |
|                                | thickness is 1.2mm , the spacing between ground copper and traces is 5.1mil         |
|                                | the board thickness is 1.6mm , the distance between ground copper and trace is 5mil |
| RF traces use 25mil line width | thickness is 1.0mm , the distance between ground copper and trace is 6.3mil         |
|                                | the board thickness is 1.2mm , the distance between ground copper and trace is 6mil |
|                                | thickness is 1.6mm , the distance between ground copper and trace is 5.7mil         |
| RF traces use 30mil line width | thickness is 1.0mm , the distance between ground copper and trace is 7.6mil         |
|                                | thickness is 1.2mm , the distance between ground copper and trace is 7.1mil         |
|                                | thickness is 1.6mm , the distance between ground copper and trace is 6.6mil         |

### 5.3.3 External Antenna

External antenna refers to the antenna that the module is installed on the outside of the product casing through IPEX extension cable, SMA and other standard RF interfaces, including rod antenna, suction cup antenna, fiberglass antenna, etc. The external antenna is basically a standard product. In order to better choose

an antenna suitable for the module, in the process of antenna selection, the parameters of the antenna should be selected as follows:

1. The working frequency of the antenna should be consistent with the working frequency of the corresponding module.
2. The input characteristic impedance of the antenna should be 50ohm.
3. The interface size of the antenna should match the size of the antenna interface of the module.
4. The standing wave ratio (VSWR) of the antenna is recommended to be less than 2, and the antenna should have a suitable frequency bandwidth ( covering the frequency points used in the actual application of specific products ) .

## **5.4 . Layout of modules**

The radiation and reception of radio frequency signals are realized by the antenna. The grounded copper sheet has a strong absorption effect on radio frequency, so the PCB onboard antenna cannot be covered by the copper sheet on the bottom plate, nor can it be covered by batteries or other metals and other devices. Surrounded, otherwise the communication distance will be greatly reduced.

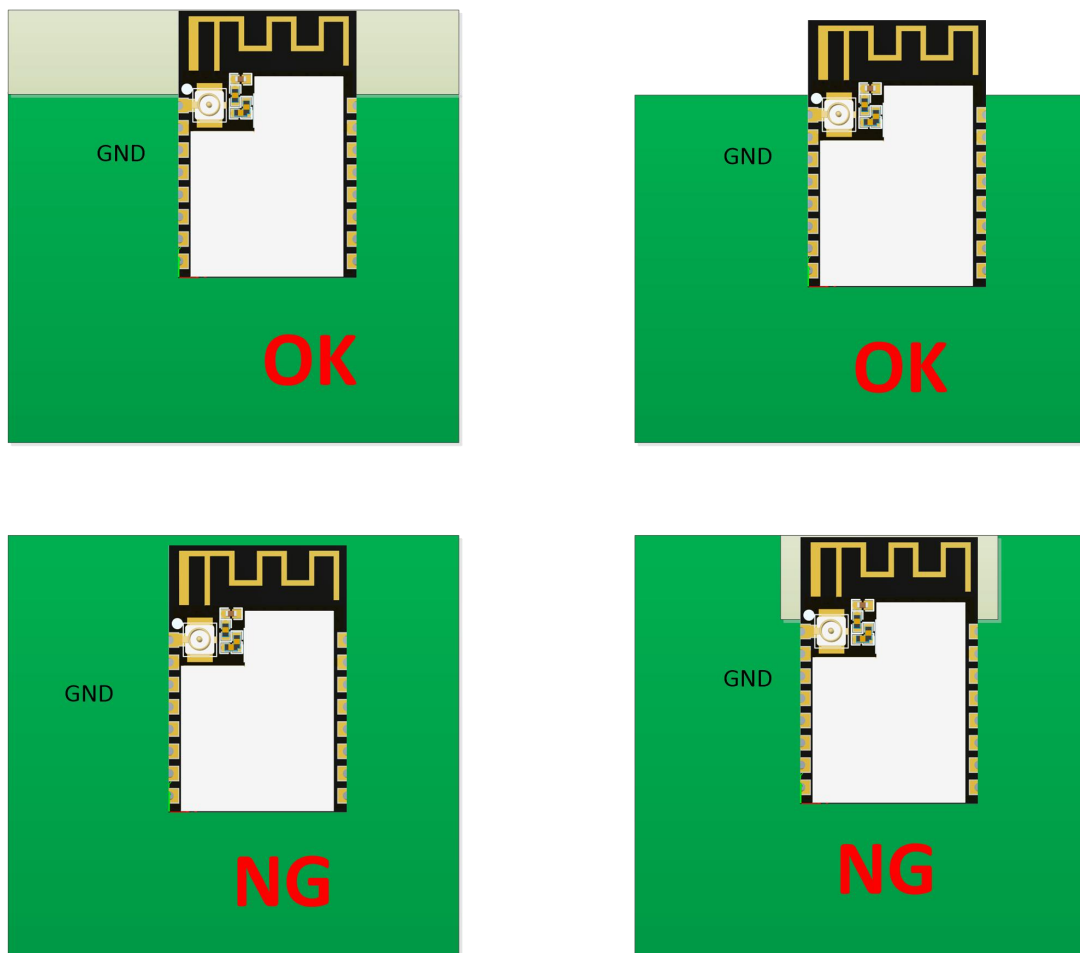


Figure 5- 5 Suggested layout of modules

## 6. Programming development

The PA&LNA power amplifier has been integrated in the module, so the output power of the chip can be set to 0dBm, and the maximum setting value should not exceed 5dBm , otherwise the PA device inside the module will be easily damaged.

```

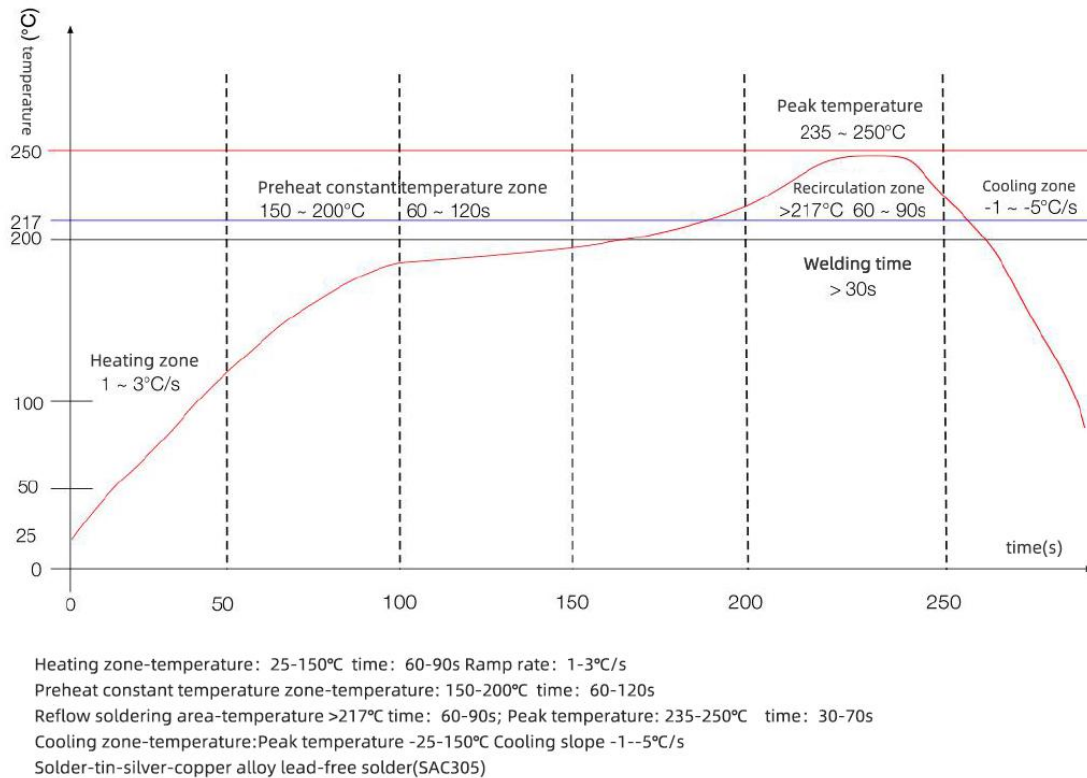
/*!
 * \brief Defines the output power in dBm
 *
 * \remark The range of the output power is [-18..+13] dBm
 */
#define TX_OUTPUT_POWER 0

```

The TXEN pin and RXEN pin of the module are the logic control pins of the PA&LNA device inside the control module. Please pay attention to the control level of the TXEN and RXEN pins of the module when using it. The control logic is listed below:

| model            | TXEN | RXEN |
|------------------|------|------|
| <b>emission</b>  | 1    | 0    |
| <b>take over</b> | 0    | 1    |
| <b>hibernate</b> | 0    | 0    |

## 7. Reflow Profile



## 8.ESD Notice

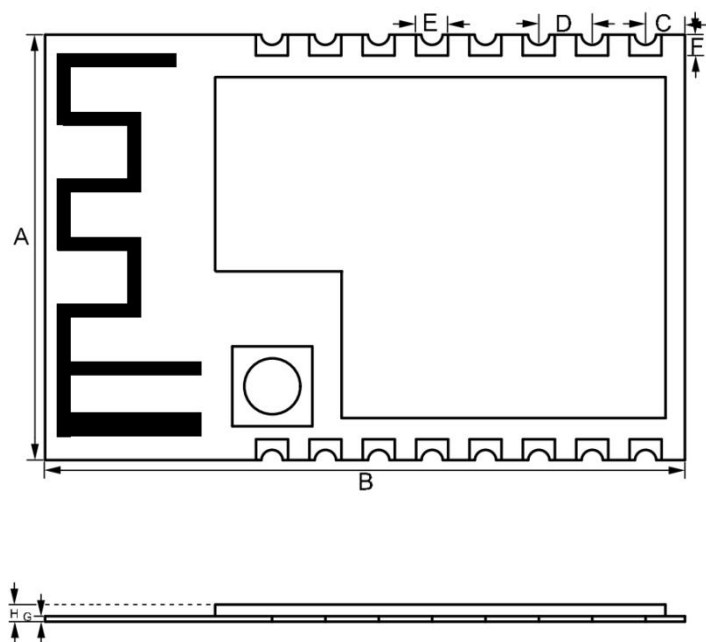
The RF module is a high-voltage electrostatic sensitive device, in order to prevent damage to the module by static electricity

- 1、 Strictly follow anti-static measures, and do not touch the module with bare hands during production.
- 2、 Modules should be placed in a placement area that can prevent static electricity.
- 3、 The anti-static protection circuit at the high voltage input should be considered in product design.



## 9. Packaging information

### Mechanical size ( unit:mm )



| Numbering | Dimensions (mm) | Error (mm) |
|-----------|-----------------|------------|
| A         | 16.0            | ±0.5       |
| B         | 24.0            | ±0.5       |
| C         | 1.46            | ±0.1       |
| D         | 2.0             | ±0.1       |
| E         | 1.2 _           | ±0.1       |
| F         | 0.6             | ±0.1       |
| G         | 1.0 _           | ±0.1       |
| H         | 2.6             | ±0.2       |



## 10. Revision History

| Revision | Comment       | Date              |
|----------|---------------|-------------------|
| V1.0     | first release | December 30, 2020 |

## 11. Ordering Information

| Index | Part Number        | Description                                                                        |
|-------|--------------------|------------------------------------------------------------------------------------|
| 1     | VG239 2 S240 X 0M2 | Tape packaging \pallet packaging<br>Factory Default PCB Onboard Antenna<br>Version |

## 12. Statement

1. Due to product version upgrades or other reasons, the content of this document will be updated from time to time. Unless otherwise agreed, this document is only used as a guide.

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